MIL-S-10539C
27 September 1978
SUPERSEDING
MIL-S-10539B
20 June 1960

MILITARY SPECIFICATION

SPECTACLES, ANAGLYPHIC, STEREO-PLOTTER, MAPPING EQUIPMENT (RED AND CYAN LENSES, WITH CASE)

This specification is approved for use by all Departments and Agencies for the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers spectacles for viewing projections in near-complementary colors made by multiplex projectors.

1.2 Classification. The spectacles shall be of the following types, as specified (see 6.2):

Type I - Temple-piece.

Type II - Clip-on.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	- Preservation-Packaging, Methods of.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The spectacles shall consist of one red and one cyan-colored lens set in a spectacle frame and constructed in such a manner that the frame may be reversed to enable the wearer to view with the red or cyan-colored lens to the left or right as desired. A pocket case shall be provided.

3.2 First article (first produced spectacles with case). The contractor shall furnish one or more spectacles with case as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design

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detail will produce spectacles with case that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).

3.4 Lenses. The lenses shall be drop shaped with the long axis not less than 46 millimeters (mm) in length and with the short axis not less than 39 mm. The edges of the lenses shall be ground and the surfaces polished. The material used for the lenses shall be of high-grade, colored glass or high-grade, laminated glass with a plastic filter medium. The lenses shall be free from striae, bubbles, stones, milkiness, lamination separation, or any other defect that would render the lens unsuitable for the use intended.

3.4.1 Red lens. The red lens shall have a refractive power of zero, plus or minus 0.12 diopter. The spectral characteristics of the red filter shall be such that transmittance of any wavelength in the visible spectrum below 560 millimicrons will not exceed 4 percent and transmittance of wavelengths above 560 millimicrons will be as great as possible. The wavelength interval on the spectrophotometric curve between the points where the transmittance is equal to 5 and 80 percent of the greatest value of transmittance in any wavelength within the visible spectrum shall not exceed 34 millimicrons, and the midpoint of this interval shall be not greater than 590 millimicrons. The average transmittance in the visible spectrum above the point on the spectrophotometric curve where the transmittance is equal to 80 percent of the greatest value of transmittance in any wavelength shall be not less than 80 percent.

3.4.2 Cyan lens. The refractive power of the cyan lens shall be minus .3750 diopter, plus or minus 0.12 diopter. The spectral characteristics of the cyan filter shall be such that the wavelength corresponding to the point on the spectrophotometric curve where the transmittance is 25 percent of the greatest value of transmittance at any wavelength will be 570 millimicrons, plus or minus 5 millimicrons. The average transmittance between 400 millimicrons and this point shall be at least 41 percent, and the average transmittance between this point and 700 millimicrons shall be not greater than 5 percent.

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3.5 Frames. The spectacle frames shall be constructed of a metal or metal alloy which has resistance to corrosion or has been treated to resist corrosion. The frames shall be sturdy to withstand normal use, and shall be adjustable without use of tools or special devices to cover a wide range of interpupillary distances and head sizes. Frames shall be designed to be comfortable to the wearer and shall be light in weight.

3.6 Case. A pocket case shall be furnished with each pair of spectacles. The case shall be resistant to attack by fungi.

3.7 Identification marking. Each pair of spectacles shall be identified in accordance with MIL-STD-130.

3.8 Type I, temple-piece. Temple pieces shall be approximately 6.5 inches (165.1 mm) overall in length.

3.9 Type II, clip-on. Spectacles shall be constructed so that they may be clipped on the lenses of ordinary spectacles.

3.10 Workmanship. The quality of workmanship shall conform to the current standards of the optical industry and shall be such as to produce spectacles that are in accordance with the requirements of this specification and are so constructed as to insure proper functioning of all parts of the units.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced spectacles with case inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced spectacles with case inspection.

4.3.1 Examination. The first produced spectacles with case shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first produced spectacles with case shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests.

4.4.3.1 Individual. Each spectacle with case shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The spectacles and case shall be examined as specified herein for the following defects:

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Major

101. Dimensions not as specified.
102. Material not as specified.
103. Frames bent or warped.
104. Lenses contain defects.
105. Case not as specified.
106. Workmanship not as specified.

Minor

201. Identification marking not as specified.

4.5.2 Tests.

4.5.2.1 Color test. The spectral characteristics of each lens shall be determined by using a spectrophotometer. Nonconformance to 3.4.1 and 3.4.2 shall constitute failure of this test.

4.5.2.2 Optical test. The refractive power of the lenses shall be determined by using a lensometer or other suitable optical instruments. Nonconformance to 3.4.1 and 3.4.2 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107. Materials, methods, or containers not as specified. Each incorrect material, method, or container shall be considered one defect.
108. Consolidation containers not closed Method V as specified.

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109. Strapping not as specified.
110. Marking illegible, incorrect, incomplete, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each pair of spectacles contained in the case shall be preserved in accordance with MIL-P-116, Method IA-8.

5.1.1.1 Consolidation. Spectacles of like description shall be consolidated together, in quantities not to exceed 25 lbs per box, in close-fitting boxes conforming to PPP-B-636, Wbc, style optional. Box closure shall be in accordance with the box specification, Method V.

5.1.2 Commercial. Spectacles with case shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

5.2.1 Level A. Spectacles of like description, preserved as specified in 5.1, shall be packed in close-fitting boxes conforming to PPP-B-601, Overseas Type, style optional. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish B. When specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for Level A except that boxes shall be Domestic Type and strapping shall be Finish A.

5.2.3 Commercial. Spectacles, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

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5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The spectacles are intended to be used to view projections in near-complementary colors made by multiplex projectors for obtaining a precise spatial stereoscopic image for use in map compilation work.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of spectacles required (see 1.2).
(c) Time frame required for submission of the first produced spectacles with case and number of spectacles with case required (see 3.2).
(d) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(e) When strapping shall be Finish A (see 5.2.1).

6.3 First produce spectacles with case. Any changes or deviations of production spectacles with cases from the approved first produced spectacle(s) with case during production will be subject to the approval of the contracting officer. Approval of the first produced spectacle(s) with case will not relieve the contractor of his obligation to furnish spectacles with case conforming to this specification.

6.4 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3)

Custodians:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE MIL-S-10539C Spectacles, Anaglyphic, Stereo-plotter Mapping Equipment (Red and Cyan Lenses, with Case)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

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